



Decision Trees

- Decision Trees: tree-based models for classification and regression, explaining structure, decision rules, and the C4.5 algorithm.
- Attribute Selection: metrics like Gini Index and Information Gain (with Gain Ratio), entropy concepts, and pruning techniques to avoid overfitting.
- Random Forests, ensemble methods (bagging, boosting, stacking), and Scikit-Learn examples with the Wine dataset.

What Are Decision Trees?



Decision Trees are intuitive models for classification and regression.

Represent decisions as a tree structure.

Internal nodes: attributes; branches: values; leaves: classes.

Decision Trees



Decision trees are one of the most simple yet widely used machine learning methodologies

- Given attribute/value **representations**
- Given a **target property to classify**
- Given a training set, inductively learn a **tree**
- Each **internal** node of the tree is an **attribute**
- Each branch is a **value** (or a range) for that attribute
- Each leaf is one of the **classes**

Decision Trees



Example – Golf Decision Tree

"Shall we play golf?"

Attributes: Outlook, Temperature, Humidity, Wind.

Target: Play or Not Play.

Decision Rules

Each path from root to leaf = a classification rule.

Example: If Outlook = Sunny AND Humidity = High → Don't Play.

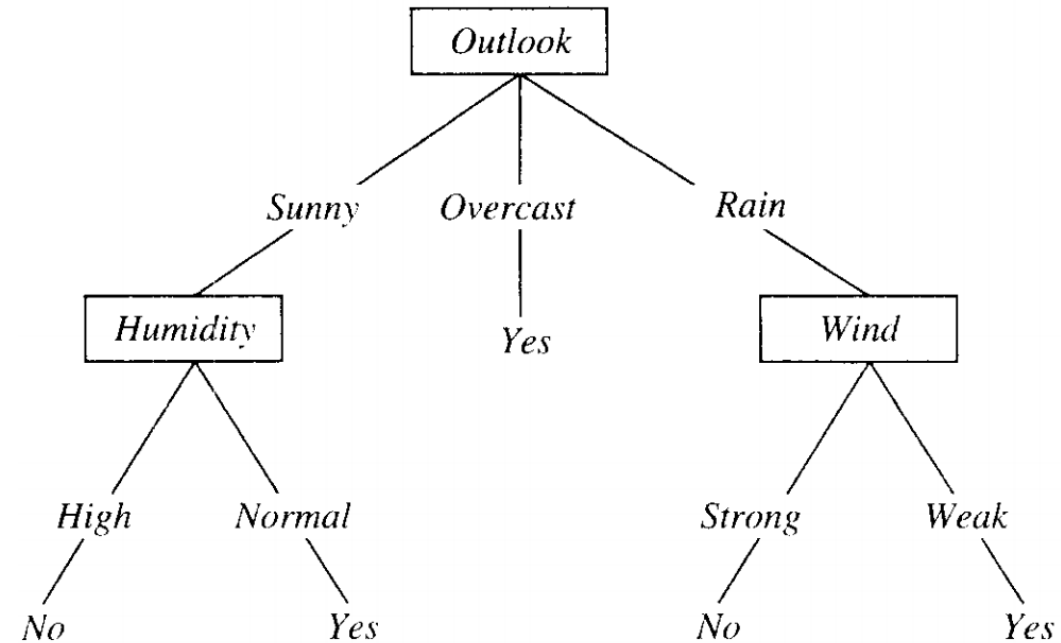
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Shall we play golf?

ID	ATTRIBUTES				TARGET
No	Outlook	Temp (° F)	Humid (%)	Windy	Class
D1	sunny	75	70	T	P
D2	sunny	80	90	T	N
D3	sunny	85	85	F	N
D4	sunny	72	95	F	N
D5	sunny	69	70	F	P
D6	overcast	72	90	T	P
D7	overcast	83	78	F	P
D8	overcast	64	65	T	P
D9	overcast	81	75	F	P
D10	rain	71	80	T	N
D11	rain	65	70	T	N
D12	rain	75	80	F	P
D13	rain	68	80	F	P
D14	rain	70	96	F	P

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Shall we play golf?



[Image by Tom Mitchell]

A path from root to leaf is a **classification rule**

Decision Trees



Many algorithms exist that **learn** decision trees
One of the most famous is C4.5 by [R. Quinlan, 93]

C4.5 Algorithm Steps

- Choose best attribute A.
- Partition dataset by A's values.
- Recursively apply C4.5 to subsets.
- Stop when subset is pure or nearly pure.

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Attribute Selection Criteria

- Perfect attribute: splits data into pure subsets.
- Useless attribute: no change in class distribution.
- Use information theory to measure effectiveness.

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Possible metrics employed for attribute selection:

- Gini Index (to minimize impurity)
- Information Gain Ratio (to maximize entropy reduction)

Decision Trees: Gini Index

Gini Index Explained

- Measures impurity of a node.
- Lower Gini = purer node.
- Used to select best attribute for splitting.

Gini Index Properties

- Value ranges from 0 (pure) to 1 (impure).
- Easy to interpret.
- Can be misleading with attributes having many values.

Why Gini Works

- Pure nodes have $Gini = 0$.
- Impure nodes have higher Gini.
- Helps identify meaningful splits.

Decision Trees: Gini Index



Gini index or Impurity measure

- measures the degree or probability of a particular variable being wrongly classified when it is randomly chosen

Impurity:

- If all the elements belong to a single class, then it can be called pure

The degree of Gini Index varies between 0 and 1:

- '0' denotes that all elements belong to a certain class or there exists only one class (pure)
- '1' denotes that the elements are randomly distributed across various classes (impure)

Decision Trees: Gini Index

Gini index (to be **minimized**)

- Perform a split: Divide node **N** into **B** child nodes based on an attribute
- **Compute Gini for the split** using:

$$G(S) = \sum_{i=1}^B \rho_i \left(1 - \sum_{k=1}^K p_{ik}^2 \right)$$

Where:

- p_i = proportion of examples from node **N** that go into child **i**
- p_{ij} = proportion of examples of class **j** in child **i**
- C = number of classes

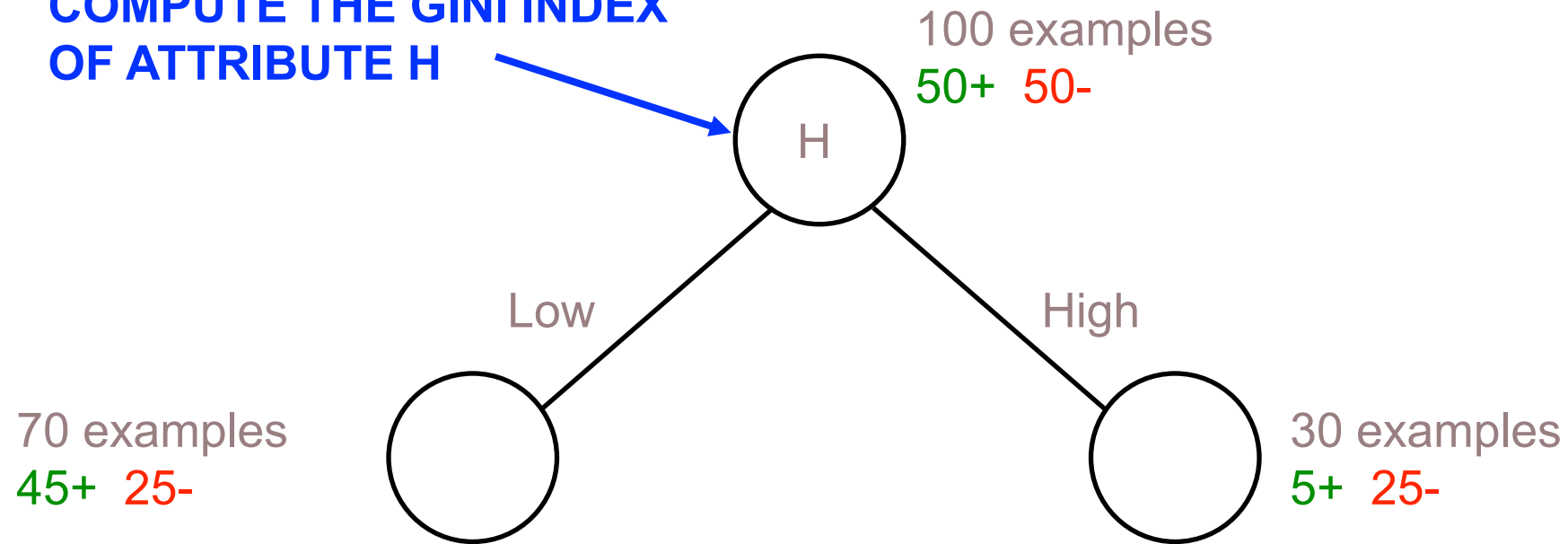
Interpretation:

- If all examples in a child node belong to one class → Gini = 0 (pure).
- If classes are evenly distributed → Gini is higher (impure).

Goal: Choose the attribute that produces the **smallest Gini index**, because it creates the purest split.

Decision Trees: Gini Index

COMPUTE THE GINI INDEX
OF ATTRIBUTE H



$$G(H) = 70/100 * (1 - (45/70)^2 - (25/70)^2) + 30/100 * (1 - (5/30)^2 - (25/30)^2)$$

Decision Trees: Gini Index



Advantage of Gini index

- Node purity is **easily interpretable**

Drawback of Gini index

- Attributes with **many values** can induce a **not meaningful** split with a very high purity

Example: fiscal code of a person

- Better to use it with attributes with **few values**

Decision Trees: Information Gain



Measures reduction in entropy after a split.
Higher gain = better attribute.

Entropy Explained

- Entropy = measure of uncertainty.
- Maximized when outcomes are equally likely.

Information Gain Ratio

- Normalizes Information Gain by attribute entropy.
- Used in C4.5 to avoid bias toward many-valued attributes.

Gain Ratio – Pros and Cons

- Corrects bias of Information Gain.
- May favor low-entropy attributes unfairly.

Decision Trees: Information Gain

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \cdot H(S_v)$$

Where:

- $H(S)$: Entropy of the original dataset S
- S_v : Subset of S after splitting by attribute A
- $|S_v| / |S|$: Proportion of examples in subset S_v
- $H(S_v)$: Entropy of subset S_v

Interpretation:

- Information Gain measures reduction in entropy after a split.
- Higher IG means better attribute for splitting.
- Goal: Maximize IG to reduce uncertainty in data.

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A note on entropy

- It measures **disorder** in data, that is the average **information** in a variable's outcome
- Highest value when **uncertainty** is highest

Example: if tossing a coin the entropy of the outcome is maximum when heads and tails are equally probable

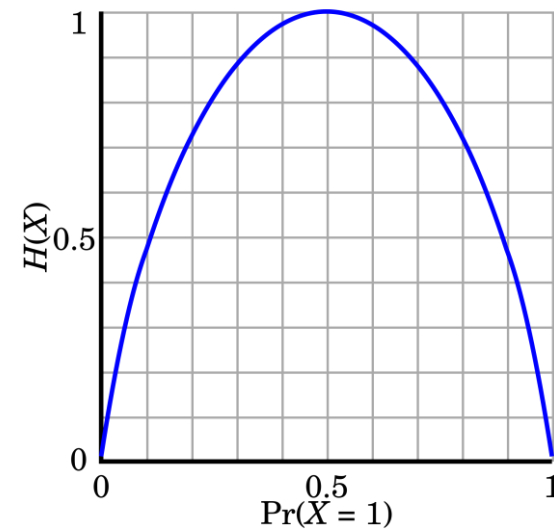
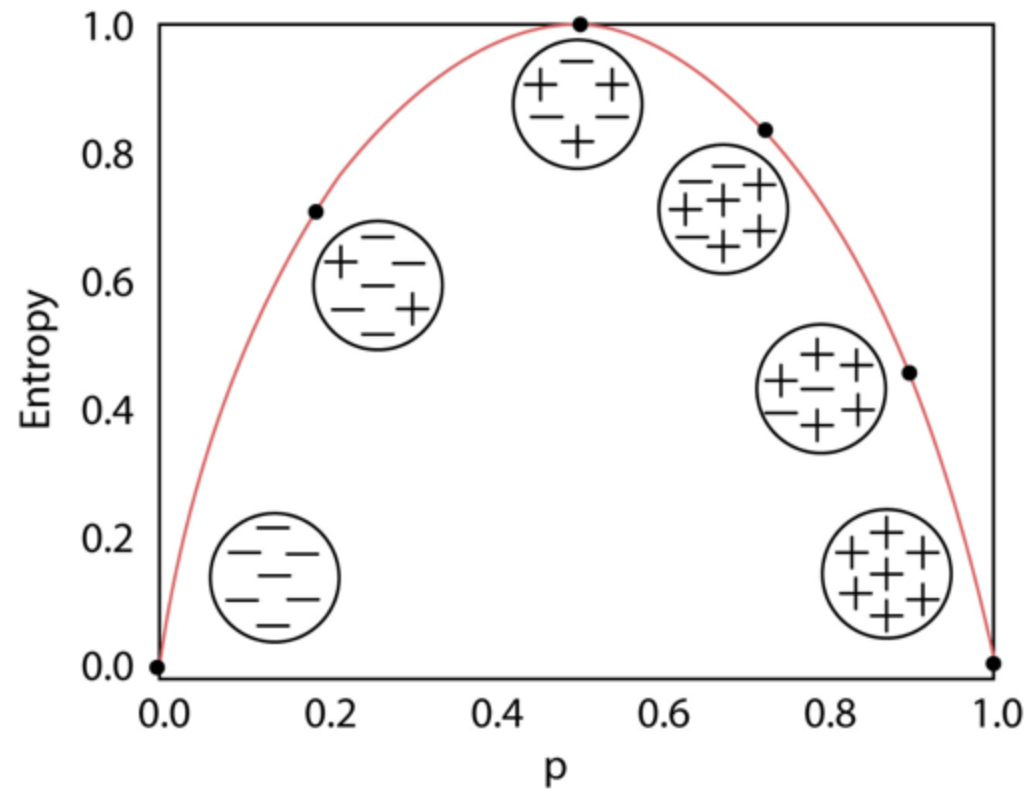


Figure from Wikipedia

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Variation of entropy against data points:



By Chainika Thakar and Shagufta Tahsildar

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Information Gain Ratio **USED IN C4.5**

- The solution is to normalize the Information Gain by the **entropy of the attribute**

$$IGR(S) = \frac{IG(S)}{H_A(N)}$$

where $H_A(S) = - \sum_{j=1}^{|A|} p_j \log_2(p_j)$

being p_j the % of examples with value j for A

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Information Gain Ratio

- **Advantage:** it solves the problem with Information Gain and Gini regarding attributes with many values
- **Disadvantage:** it may give advantage to attributes having a low information gain (just because they have a low entropy)

Decision Trees



Decision trees can easily **overfit** data

- **Pre-pruning:** Stop growing tree when:
 - Node is pure or nearly pure.
 - Few examples remain.
 - No gain in purity.
- **Post-pruning:** Build full tree, then prune:
 - Remove subtrees that don't improve validation accuracy.
 - Prefer simpler trees (Occam's Razor).

Decision Trees



How to handle continuous variables

- Try all possible split values.
- Sort values to improve efficiency.
- Apply same metrics (Gini, Gain).

How to handle missing values...

- Ignore missing values in metric computation.
- Assign examples to majority branch or weight all branches.

Decision Trees



Pros

- Simplicity
- Interpretability — **glass box**
- Can handle **missing values**
- Can handle **continuous and categorical** variables

Cons

- Highly prone to **overfitting**
- High variance with **small changes** in training sets

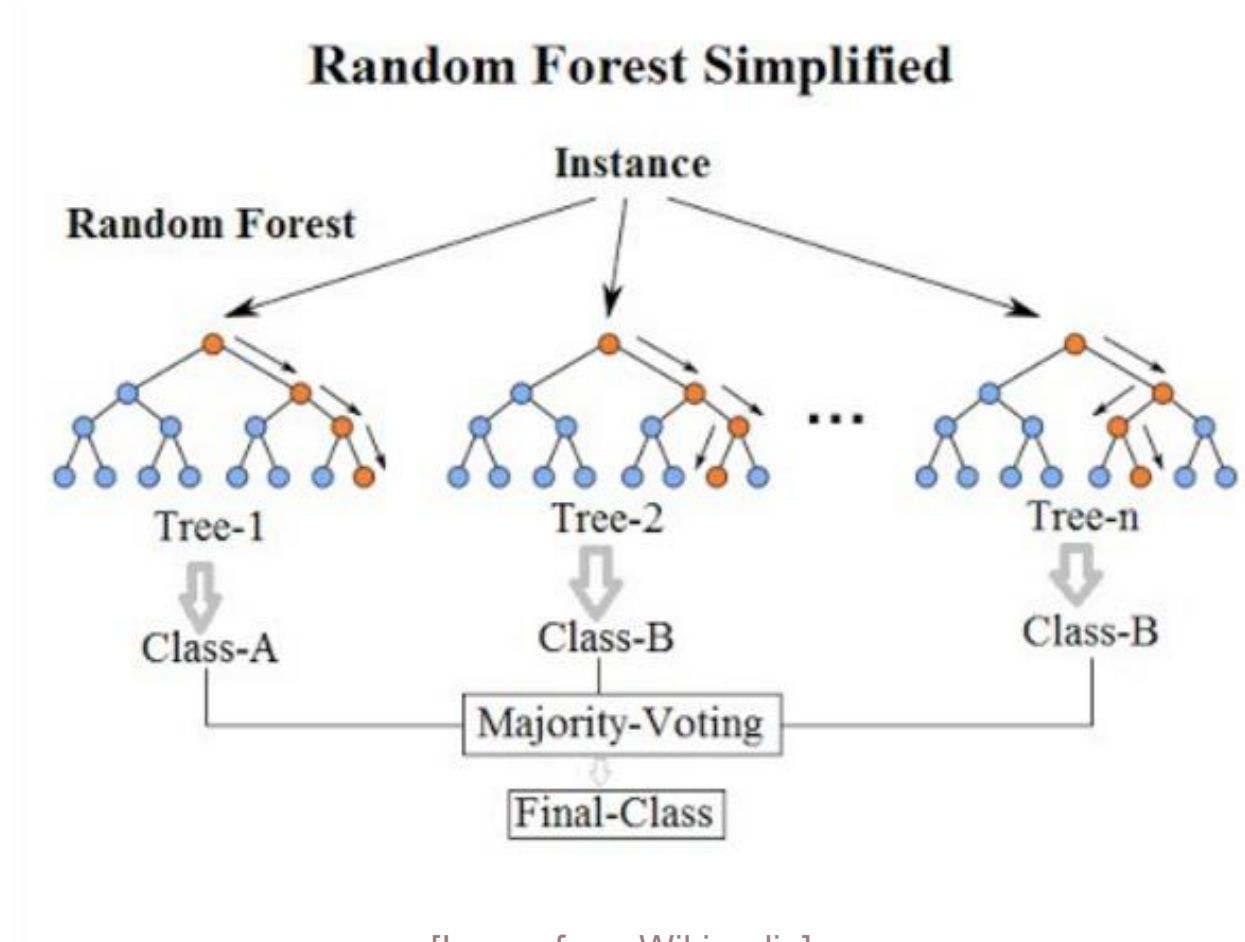
Decision Trees: Random Forests



A very effective extension: **Random Forests**

- Many **different** decision trees
- Each learned on a **subset** of the attributes
- **Average** prediction over all trees
- Named an **ensemble** method
- Typically **higher performance** than decision trees

Random Forests



[Image from Wikipedia]

Ensemble Methods



Combining weak classifiers into stronger ones: often this brings an improvement in terms of performance

- Bagging
- Boosting
- Stacking

Ensemble Methods



Bagging

- Homogeneous weak learners
- Learning in parallel
- Somehow aggregate the results

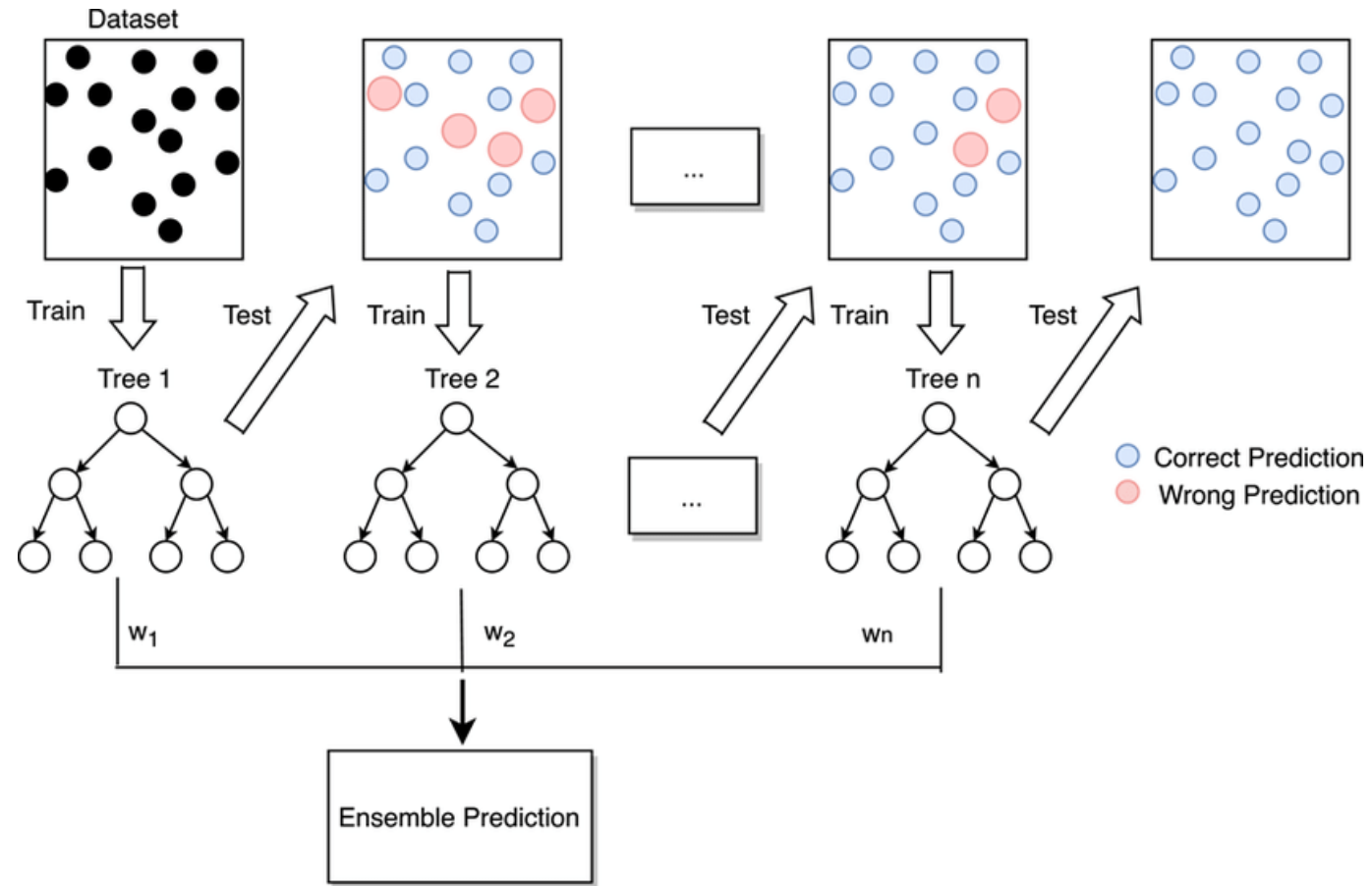
Ensemble Methods



Boosting

- Homogeneous weak learners
- Learning sequentially
- Combination with some deterministic strategy

Gradient Boosting



[Image from Zhang et al.]

Ensemble Methods



Stacking

- Heterogeneous weak learners
- Learning in parallel
- Feed predictions of several models into a higher level model (meta-model)

Regression Trees



An **extension** of decision trees to regression

- Use the **improvement** in Mean Squared Error to evaluate and select attributes during training
- In a leaf node, compute the **average** of the target variable across the training examples in the leaf
- Use **ensemble** techniques (Random Forests and Gradient Boosting) as for classification